

MATH0075 Lie Groups and Lie Algebras

<i>Year:</i>	2025–2026
<i>Code:</i>	MATH0075
<i>Level:</i>	7 (UG)
<i>Value:</i>	15 units (= 7.5 ECTS credits)
<i>Normal student group(s):</i>	UG Year 3 and 4 Mathematics degrees PG: (2026-27) MSc Mathematics (level 7)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0014, MATH0053
<i>Lecturer:</i>	Dr Y Naqvi

Course Description and Objectives

Lie groups are continuous groups of symmetries, like the group of rotations of n -dimensional space or the group of invertible n -by- n matrices. In studying such groups we can use tools from calculus to linearise our problems, which leads us to the notion of a Lie algebra: a vector space with an antisymmetric product associated to any Lie group, which remembers everything about its algebraic structure. For example, the Lie algebra associated with the group of rotations of 3-space is just 3-dimensional Euclidean space with (twice) the vector cross product.

This course divides in two halves. In the first half we introduce the notion of a Lie algebra and the relationship between a Lie group and its Lie algebra. This will involve some ideas from geometry (manifolds and tangent spaces) which will serve you well in later courses. In the second half we study representations of Lie groups and Lie algebras, paying attention to the groups $SU(2)$ and $SU(3)$. This will be much more algebraic.

Recommended Texts

- (i) B. C. Hall, *Lie groups, Lie algebras and representations*, Springer GTM, 222, 2015.
- (ii) R. Carter, G. Segal, I. McDonald, *Lectures on Lie groups and Lie algebras*, LMS Student Texts, 32, CUP 1995.
- (iii) W. Fulton, J. Harris, *Representation theory: a first course*, Springer GTM, 129, 1991.
- (iv) A. Kirillov, Jr., *An introduction to Lie groups and Lie algebras*, Cambridge Studies in Advanced Mathematics, 113, CUP 2008.
- (v) F. Warner, *Foundations of differentiable manifolds and Lie groups*, Springer GTM, 94, 1983.

Detailed Syllabus

1. From Lie groups to Lie algebras

- Lie groups: definition and examples.
- The adjoint representation.
- The exponential map, Lie bracket and Lie algebras.

- Lie's theorems.

2. Representations

- Representations of Lie groups and Lie algebras.
- Abelian Lie groups.
- Representations of $SU(2)$ and $SO(3)$.
- Representations of $SU(3)$.
- Sketch of the classification of simple Lie algebras.